MOHAMMED ALA EDDINE BENHAMMOU

Senior Odoo Full-Stack Developer & ERP Solutions Architect

Email: b.aladinmed@gmail.com
Phone: +974 31410671 | **Location:** Doha, Qatar
GitHub: github.com/aladinbndhh
Portfolio: aladinbndhh.vercel.app

EXECUTIVE SUMMARY

High-impact **Senior Odoo Full-Stack Developer & Solutions Architect** with extensive production experience engineering **599+ custom modules** across **Odoo 14.0 to 18.0** at an **Odoo Gold Partner** (NextMove Solutions). Specialized in Python backend ORM architecture, OWL 2.0 (Odoo Web Library) frontend components, high-throughput PostgreSQL query optimization, and physical hardware-to-cloud bridges (automated tuition payment kiosks, venue POS terminals, BOCA thermal printers, BioTime biometrics). Proven track record leading complex enterprise ERP and SIS digital transformations for universities, government agencies, and retail enterprises. Trilingual professional (English, French, Arabic) with Master's coursework in distributed systems.

ODOO CORE COMPETENCIES & TECHNICAL SKILLS

Odoo Framework:	**Odoo 18.0, 17.0, 16.0, 15.0, 14.0** | ORM Internals (Recordsets, `self.env`, prefetching, cache, computed fields, `@api.depends`, `@api.constrains`), Model Inheritance (`_inherit`, `_inherits`), Abstract/Transient Models, Custom Wizards.
Frontend & UI:	**OWL 2.0 (Odoo Web Library)**, QWeb XML Templates (Form, Tree, Kanban, Pivot, Graph), Custom OWL Widgets, POS Web Client Extensions, JavaScript ES6+, HTML5/CSS3.
Backend & APIs:	**Python 3**, JSON-RPC, XML-RPC, RESTful Webhooks, Custom Controllers (`http.route`), Ellucian Banner SIS Integration, Payment Gateways (QPay), External API Aggregators.
Database & Performance:	**PostgreSQL** (Query Profiling with `EXPLAIN ANALYZE`, B-Tree Indexing, Table Partitioning, Connection Pooling, Elimination of $N+1$ query bottlenecks in ORM loops).
Hardware / IoT Integration:	Interactive Student Payment Kiosks (NFC/Card clearance), BOCA Thermal Ticket Printers (custom FGL socket daemon), ZKTeco BioTime Biometric Attendance, POS Barcode Peripherals.
DevOps & Tooling:	Linux (Ubuntu/Debian Server), Docker, Docker-Compose, Nginx Reverse Proxy, Git / GitHub CI/CD, Automated Testing, Multi-Instance Deployments.

PROFESSIONAL EXPERIENCE

Senior Full-Stack Developer & Enterprise Solutions Architect | NextMove Solutions (Odoo Gold Partner)

2023 — Present | Doha, Qatar

- **Ecosystem Leadership & Module Delivery:** Architected, developed, and maintained over **599 custom enterprise Odoo modules** spanning Accounting (109), Human Resources (136), POS, Logistics, and Localizations across major enterprise client accounts.
- **Lusail University — Campus ERP & SIS Kiosk Integration:** Architected bi-directional Odoo ERP synchronization with **Ellucian Banner SIS** for student tuition billing. Engineered end-to-end self-service physical tuition payment kiosks handling local NFC card drivers, debit gateway clearance, and atomic ledger reconciliation. Deployed 75+ custom academic modules including faculty payroll, multi-tiered financial approval hierarchies, and digital document signing.
- **Qatar Cinema — Multi-Branch Venue POS & Thermal Ticket Printing:** Spearheaded full Odoo 17 & 18 migration for commercial cinema branches. Developed a high-performance Python socket server (`boca_printer_server.py`) converting POS transactions into native FGL bytecode for BOCA thermal printers. Designed real-time Cinema API gateways for seat map allocation, box-office synchronization, and QTickets third-party aggregator integration.
- **National Cyber Security Agency (NCSA) — Security Architecture:** Hardened Odoo platform adhering to national cybersecurity zero-trust frameworks. Enforced fine-grained Role-Based Access Control (RBAC), multi-factor authentication, cryptographic audit logs, and automated PostgreSQL encrypted backup/failover pipelines.
- **Qinwan Dates & Tajer — Luxury Retail & Supply Chain:** Engineered custom e-commerce product configurators in OWL, dynamic promotional coupon engines, multi-branch inventory routing, and automated DHL international logistics integration.
- **Database Optimization & Code Governance:** Conducted code reviews for junior developers, enforced strict PEP 8 and Odoo design guidelines, and tuned high-traffic PostgreSQL queries to eliminate bottlenecks and optimize memory prefetching.

EDUCATION & ACADEMICS

Master's Degree Coursework in Computer Science
Advanced Distributed Systems & Software Engineering (3 Semesters Completed)

Bachelor of Science (B.Sc.) in Computer Science
Focus on Systems, Networks & Software Architecture (3-Year Program)

LANGUAGES & COMMUNICATION

French: Fluent / Bilingual Proficiency (C2 — Professional & Technical)

English: Fluent / Professional & Technical Proficiency (Full Working)

Arabic: Native Speaker

Experienced in multicultural client environments & technical leadership